EXISTENCE OF A GLOBALLY GENERATED INVERTIBLE SHEAF WITH PRESCRIBED NONVANISHING LOCUS ON AN INTEGRAL SEPARATED SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Existence of an element whose inverse lies in the Jacobson radicals of the local rings at the missing points. Stacks Project, Varieties, Lemma lemma-create-globally-generated (tag 09N5). Supplies the element f from which the cocycle for the line bundle is built.

Source reference. Original lemma 09N5: sources/stacks-chow/tags.txt, source lines 12203–12203. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be an integral separated scheme. Let $U \subset X$ be a nonempty affine open such that $X \setminus U$ is a finite set of points $x_1, \ldots, x_r$ with $\mathcal{O}_{X,x_i}$ Noetherian of dimension 1. Then there exists a globally generated invertible $\mathcal{O}_X$-module $\mathcal{L}$ and a section s such that $U = X_s$.*

Proof. Say $U = \mathrm{Spec}(A)$ and let K be the function field of X. Write $B_i = \mathcal{O}_{X,x_i}$ and $\mathfrak{m}_i = \mathfrak{m}_{x_i}$. Since $x_i \notin U$ we see that the open $U \times_X \mathrm{Spec}(B_i)$ of $\mathrm{Spec}(B_i)$ has only one point, i.e., $U \times_X \mathrm{Spec}(B_i) = \mathrm{Spec}(K)$. Since X is separated, we find that $\mathrm{Spec}(K)$ is a closed subscheme of $U \times \mathrm{Spec}(B_i)$, i.e., the map $A \otimes B_i \to K$ is a surjection. By Lemma 09N5 we can find a nonzero $f \in A$ such that $f^{-1} \in \mathfrak{m}_i$ for $i = 1, \ldots, r$. Pick opens $x_i \in U_i \subset X$ such that $f^{-1} \in \mathcal{O}(U_i)$. Then

$$\mathcal{U} : X = U \cup \bigcup U_i$$

is an open covering of X. Consider the 2-cocycle with values in $\mathcal{O}_X^*$ given by f on $U \cap U_i$ and by 1 on $U_i \cap U_j$. This defines a line bundle $\mathcal{L}$ with two sections:

(1) a section s defined by 1 on U and f^{-1} on U_i is as in the statement of the lemma, and
(2) a section t defined by f on U and 1 on U_i.

Note that $X_t \supset U_1 \cup \ldots \cup U_r$. Hence s, t generate $\mathcal{L}$ and the lemma is proved. □

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